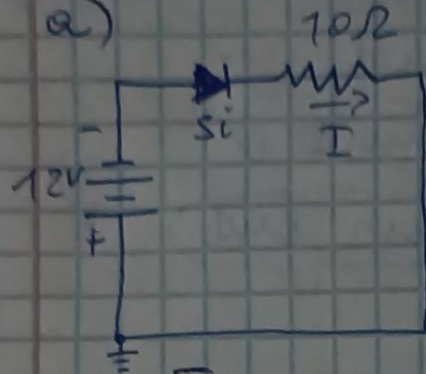


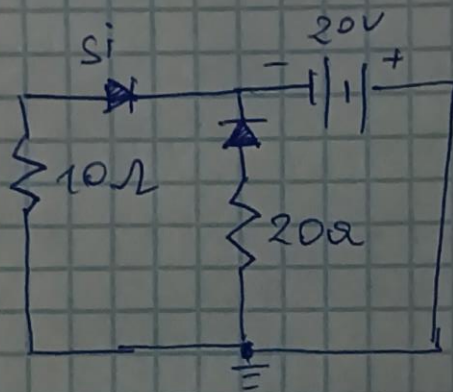
1

page 12, 3, 5

a)

 $\bar{I} = 0$  reverse-biased diode

b)



$$V = 20V - 0,7V = 19,3V$$

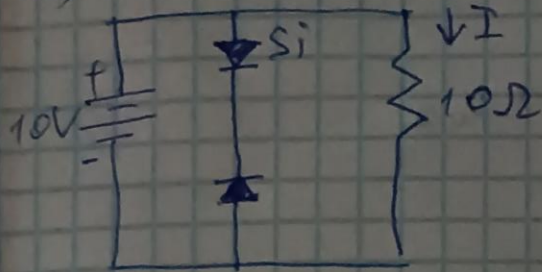
$$I(20\Omega) = \frac{19,3V}{20\Omega} = 0,965A$$

$$V(10\Omega) = 20V - 0,7V = 19,3V$$

$$I(10\Omega) = \frac{19,3V}{10\Omega} = 1,93A$$

$$I = I(10\Omega) + I(20\Omega) = 2,895A$$

c)

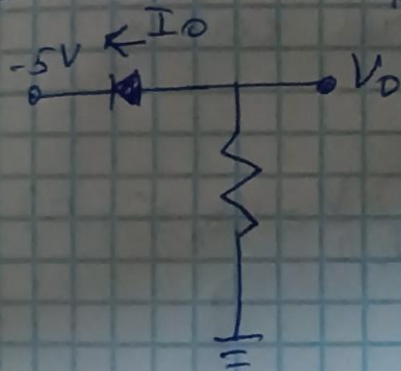


2

$$I = \frac{10V}{10\Omega} = 1A$$

center branch open

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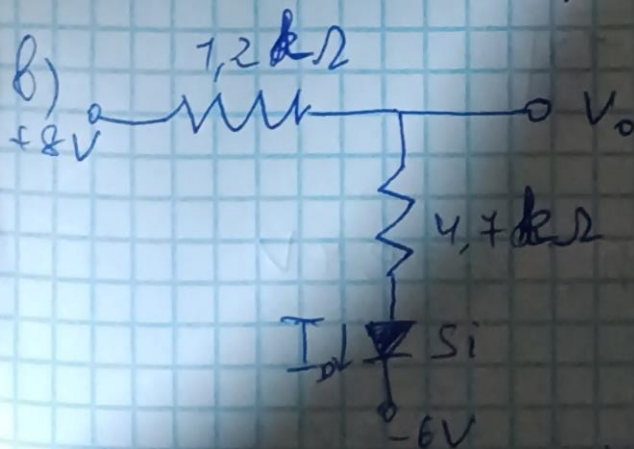


Diode forward-based Kirchhoff's voltage law:

$$-5V + 0,7V - V_0 = 0$$

$$V_0 = -4,3V$$

$$I_R = I_D = \frac{|V_0|}{R} = \frac{4,3V}{2,2k\Omega} = 1,855mA$$

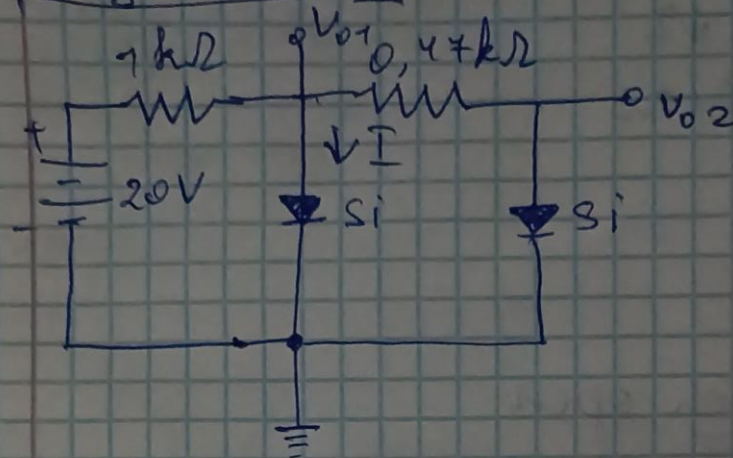


Diode forward-based

$$I_D = \frac{8V + 6V - 0,7V}{1,2k\Omega + 4,7k\Omega} = 2,25mA$$

$$V_0 = 8V - (2,25mA)(1,2k\Omega) = 5,3V$$

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3

Both diodes forward biased

$$V_{01} = 0,7V, V_{02} = 0,7V$$

$$I_{1k\Omega} = \frac{20V - 0,7V}{1k\Omega} = \frac{19,3V}{1k\Omega} = 19,3(mA)$$

$$I_{0,47k\Omega} = 0(mA)$$

$$I = I_1 - I_2 = 19,3(mA)$$

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$$16) V_{in} = 0$$

both diodes on.  $V_o = 0,7V$

$$17) V_{in} = 10V$$

both diodes off.  $V_o = 0,7V$

